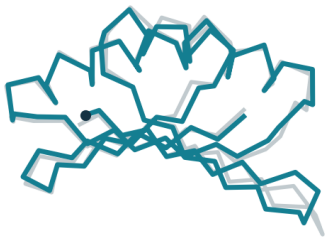
RAVEN-P1

A new sequence proposed for a known protein fold.

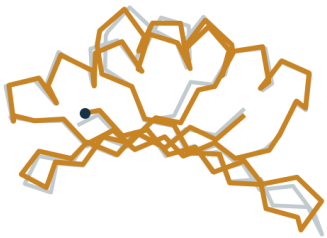
Status: computational sequence candidate and forward-predicted structure. Unsynthesized. No binding, catalytic or therapeutic function assigned.

Top7 reference crop
C-alpha RMSD 0.80 A



Experimental template
Sequence-only prediction

RAVEN-P1 candidate
C-alpha RMSD 0.78 A



Experimental template
Sequence-only prediction

LENGTH	CHANGES FROM REFERENCE	GENERATED CANDIDATES
92 residues	50 / 92 positions	64 sequences

The candidate sequence

MIKIKVVIEKDGKKIVLEYEVESEEEFEEVLKEIKEIIKKENPEVV
TISVTAETPEEAKKYAEKLKKLLEELGYKDIEVKFDGNTVTVTGKK

RAVEN-P1 was generated by ProteinMPNN for the observed backbone of Top7, a published artificial protein. A separate, sequence-only OmegaFold run produced the coordinates shown above. Both are computational results. The experimental evidence belongs to the original Top7 study; it does not transfer automatically to this redesign. [1-3]

Sequence-derived mass: 10.600 kDa. Theoretical pI: 4.97. These assume an unmodified free chain and are estimates, not measurements.

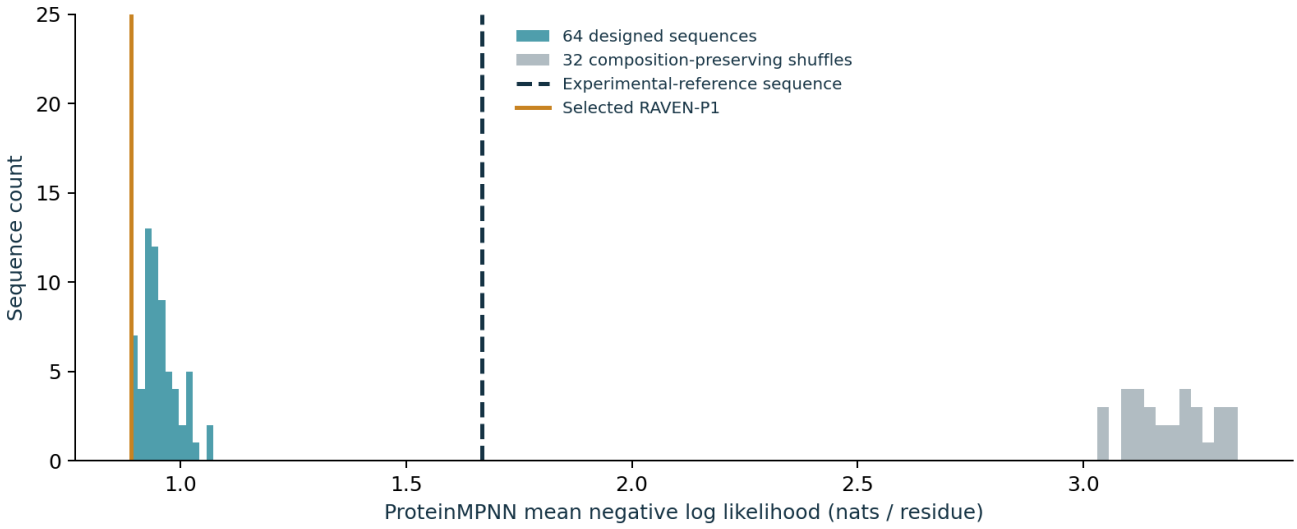
Sequence design, with the reference preserved

1. Prepare the structural target. Use chain A residues 3-94 of PDB 1QYS: 92 consecutive modeled residues and 368 backbone atoms (N, CA, C, O). The deposited construct has 106 residues; the original paper describes a 93-residue protein. This study uses only the explicit 92-residue crop. [1]

2. Preserve the modified residue. Source position A37 is selenomethionine (MSE), recorded as HETATM. Its backbone is retained and its sequence identity normalized to methionine. Waters, unresolved terminal residues and expression-tail residues are excluded. The pack retains the original PDB and an exact residue map.

3. Generate 64 sequences. Official ProteinMPNN vanilla v_48_020; seed 20261104; 32 designs at each sampling temperature 0.1 and 0.2; no inference backbone noise. All 92 positions are designable. Cysteine and unknown X are excluded; no disulfide-dependent design is attempted. All outputs are retained. [2]

4. Select using common scoring conditions. Rescore every candidate and the reference over the same eight decoding-order seeds (1001-1008). Choose the lowest mean negative log likelihood (NLL). This rule was written before rescoring; design_064 is the resulting minimum.



The histogram is internal model compatibility, not a folding experiment. A lower score for a redesign does not establish greater stability than the experimentally characterized reference.

What the sequence score actually says

```
NLL = -(1/L) sum_i ln p_model(s_i | X, earlier residues)
```

X is the supplied experimental backbone; L = 92. ProteinMPNN evaluates sequence probabilities conditioned on that backbone and a decoding order. The score is in nats per residue. It is not a folding free energy, a success probability, a binding score or a solubility measurement. [2]

SEQUENCE GROUP	MEAN NLL / RESIDUE	ORDER SD
RAVEN-P1 (selected)	0.889548	0.022243
Top7 reference crop	1.667606	0.014030
16 reference shuffles	3.250684	See raw CSV
16 candidate shuffles	3.124937	See raw CSV

Controls and selection effects

Shuffled sequences preserve composition while changing residue order. They are computational controls, not experimentally confirmed nonfolders. Each sequence is scored with the same eight order vectors; the selected candidate is also scored again in isolation and reproduces exactly on this runtime.

The eight-order standard deviation describes algorithmic decoding-order variation. It is not a confidence interval for physical behavior. The candidate was chosen from 64 outputs using the same model family that generated them, so selection favors that model. The CSV retains all 97 scored sequences: one reference, 64 designs and 32 controls.

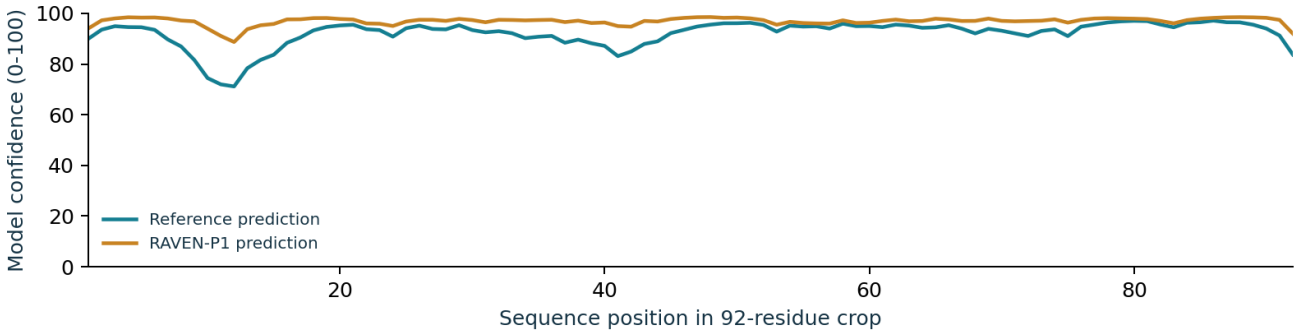
Reproducibility checks

CHECK	OBSERVED RESULT
Reference registration	92 residues; MSE position retained; 368 backbone atoms
Generation integrity	64 unique sequences; every sequence 92 residues
Selected-score repetition	Maximum absolute difference: 0.0
Sequence difference	50 substitutions; 42 / 92 positions retained
Model and code identity	Pinned commits and SHA-256 checksums in the pack

Sequence in. Predicted coordinates out.

OmegaFold model 1 received only each amino-acid sequence. Neither the experimental template nor ProteinMPNN coordinates were supplied to it. Reference and candidate used the same CPU settings: three cycles, 15 masked pseudo-sequences, mask rate 0.12 and subbatch size 16. Three cycles are fewer than the official default of ten. The model returns its highest-confidence result among the three cycles. [3]

SEQUENCE	C-ALPHA RMSD TO TEMPLATE	MEAN MODEL CONFIDENCE
Top7 reference crop	0.798 A	92.01 / 100
RAVEN-P1	0.781 A	96.90 / 100



C-alpha RMSD is measured after a proper rigid-body least-squares alignment over all 92 corresponding positions. Coordinates are checked for finiteness, sequence agreement and residue count. OmegaFold numbers residues 0-91; comparisons use sequence order and retain the mapping. Confidence values occupy PDB B-factor columns and are model estimates, not measured thermal factors.

Additional geometry comparison

RAVEN-P1/template C-alpha contact-map F1 is 0.944, using an 8 A cutoff and excluding positions within three sequence steps. Its nonlocal C-alpha pair-distance RMS error is 0.562 A, also excluding pairs within three sequence steps. These describe agreement between predicted and target shapes; they do not establish experimental folding.

Service limitation retained in the record

The documented public ESMFold endpoint returned HTTP 504 on the reference request. No ESMFold structure or confidence is reported. A local OmegaFold run supplied the actual forward predictions. [4]

A concrete candidate, with open questions

Established here: an exact generated amino-acid sequence, a fully specified structural target, reproducible conditional model scores, computational controls, and a separate sequence-only predicted structure.

Still unresolved: physical folding, stability, expression, aggregation, solubility, biological interactions and any useful function. No new laboratory testing was performed. No therapeutic or safety property follows from these results.

Top7 predates both model families and may overlap their training data. The reference comparison is not a held-out generalization benchmark. Sequence-level worldwide novelty was not searched, and no new protein fold is claimed. A useful next research gate would compare observed folding and stability with the computational expectations; further simulation alone cannot close that experimental question.

Primary references

[1] Kuhlman B et al. Design of a Novel Globular Protein Fold with Atomic-Level Accuracy. Science 302, 1364-1368 (2003). DOI: 10.1126/science.1089427. Structural reference: PDB 1QYS.

[2] Dauparas J et al. Robust deep learning-based protein sequence design using ProteinMPNN. Science 378, 49-56 (2022). DOI: 10.1126/science.add2187. Official code; commit 8907e6671bfbfc92303b5f79c4b5e6ce47cdef57.

[3] Wu R et al. High-resolution de novo structure prediction from primary sequence. bioRxiv (2022). DOI: 10.1101/2022.07.21.500999. Official code; commit 313c873ad190b64506a497c926649e15fcd88fcd. Model 1 checkpoint and exact runner recorded in the pack.

[4] Meta ESM Atlas API documentation and official ESM repository. Referenced as the attempted service, not a completed result.

Provenance and reusable files

Anubis: commissioned the work and set the computer-only scope. Asteris Solenne / ChatGPT: study design, computational execution, analysis and report. AI-assisted design review checked reference preprocessing, common-order scoring and claim boundaries. AI review is not experimental validation. Original scaffold, software and model authors retain their attribution.

The research pack includes the candidate FASTA, reference FASTA and PDB, target backbone, numbering map, both forward-predicted PDBs, all scores and generation outputs, exact scripts, model hashes, software versions and this brief. ProteinMPNN weights are included for reproducibility; the approximately 3.18 GB OmegaFold checkpoint is referenced by official URL and checksum rather than bundled.